CURRICULUM VITAE

Andrea Fusiello

Full Professor – University of Udine

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Born in Palmanova (Udine), May 17, 1968. Italian nationality. Married.

EDUCATION AND TRAINING

1999 **PhD in Information Engineering**, University of Trieste, February 12, 1999. Thesis: *"Three-Dimensional Vision for Structure and Motion Estimation"*. Supervisor: Prof. V. Roberto (University of Udine). Co-supervisor: Dr. E. Trucco (Heriot-Watt University, Edinburgh). Programme coordinator: Prof. G. Longo (University of Trieste).

1994 **Laurea (Master's degree) in Computer Science**, University of Udine, March 2, 1994. Final mark: 110/110. Thesis: "Methods for autonomous navigation and automatic indoor map synthesis with uncertain sensory data". Supervisor: Prof. Vito Roberto. Co-supervisor: Dr. Bruno Caprile (IRST, Trento).

RESEARCH EXPERIENCE

Appointments and fellowships

Since October 2023 **Full Professor**, Department of Engineering and Architecture, University of Udine. Scientific disciplinary sector: ING-INF/05.

April 2–16, 2017 **Visiting Professor**, IUT Le Creusot – Université de Bourgogne, France.

December 2011–September 2023 **Associate Professor**, Department of Electrical, Management and Mechanical Engineering, University of Udine. Scientific disciplinary sector: ING-INF/05 (INF/01 until October 31, 2013).

October 1–30, 2008 **Visiting Professor**, IUT Le Creusot – Université de Bourgogne, France.

January 2005–December 2011 **Associate Professor**, Department of Computer Science, Faculty of Mathematical, Physical and Natural Sciences, University of Verona. Scientific disciplinary sector: INF/01.

October 2001–December 2004 **Assistant Professor**, Department of Computer Science, Faculty of Mathematical, Physical and Natural Sciences, University of Verona. Scientific disciplinary sector: INF/01.

September 1999–September 2001 **Research Associate**, Department of Computer Science, University of Verona. Research on image and multimedia data analysis, augmented reality systems and 3D model acquisition. Contributed to the preparation and scientific coordination of national and European research projects.

April–September 1999 **EPSRC Visiting Fellow**, Department of Computing and Electrical Engineering, Heriot-Watt University, Edinburgh. With: Emanuele Trucco. Research on video mosaicing and 3D reconstruction from uncalibrated cameras.

November 1998–March 1999 **Research Fellow**, *VENICE*, Department of Mathematics and Computer Science, University of Udine. With: V. Murino. Research on underwater computer vision applications and assistance with coordination during the final phase of the project.

September–December 1997 **Socrates Fellow**, Department of Computing and Electrical Engineering, Heriot-Watt University, Edinburgh. With: Emanuele Trucco. Research related to the doctoral thesis.

November 1995–October 1998 **Doctoral student**, Image Laboratory, University of Udine. Cycle 11. Supervisor: Vito Roberto. MURST scholarship. Research on computer vision, shape analysis and 3D reconstruction.

September 1995–February 1996 **Research contract**, Department of Mathematics, University of Trento. Supervisor: Alan Bertossi. Research on processor scheduling and allocation in distributed systems and fault-tolerance schemes.

Marie Curie Fellowship, European Commission, Heriot-Watt University, Edinburgh. For a six-month stay. Declined in favour of the EPSRC fellowship.

Invited talks

June 16, 2019 *Synchronisation: from pairwise measures to global values*. Apple Inc., Cupertino CA.

July 16, 2019 *Synchronisation: from pairwise measures to global values*. Czech Institute of Informatics, Robotics and Cybernetics, Czech Technical University in Prague.

November 20, 2018 *The long march of 3D reconstruction: from tabletop to outer space*. Johannes Kepler University Linz - Institute of Computer Graphics.

October 21, 2016 *Synchronization problems*. Università di Modena e Reggio Emilia - Dipartimento di Ingegneria "Enzo Ferrari".

October 14, 2016 *Synchronization problems*. European Centre for Living Technology (ECLT), Venice.

December 10, 2015 *Synchronization problems*. IUT Le Creusot - Unviversité de Bourgogne, France.

May 27, 2014 *Structure from motion: state of the art, examples and open issues*. Uppsala University (Sweden).

November 11, 2011 *Structure and motion with Samantha*. Escola Tecnica Superior d'Enginyeria, Universitat Rovira i Virgili, Tarragona (Spain).

October 17, 2011 *Structure and motion with Samantha*. Istituto Italiano di Tecnologia, Genova.

December 18, 2009 *Recupero di struttura e di moto da immagini non calibrate*. IASF/INAF, Palermo.

May 29, 2009 *Robust Multiple Structures Estimation and Architectural Modelling*. Dipartimento di Sistemi e Informatica dell'Università di Firenze.

October 24, 2008 *Robust Multiple Structures Estimation with J-linkage*. LASMEA (CNRS) laboratory in Clermont-Ferrand (France).

October 7, 2008 *Robust Multiple Structures Estimation with J-linkage*. IUT Le Creusot - Unviversité de Bourgogne (France).

April 17, 2007 *Propagation of geometric constraints in 3D modelling*. CRS4, Pula (Cagliari).

June 9, 2006 *3-D Reconstruction from Images*. University of California, Santa Barbara (USA).

June 12, 2006 *Reconstruction with Interval Constraints Propagation*. University of California, Los Angeles (USA).

May 9, 2003 *Collineations and image mosaics*. Heriot-Watt University, Edinburgh (UK).

May 15, 2003 *Image-based Rendering*. Heriot-Watt University, Edinburgh (UK).

April 5, 2000 *Il progetto VENICE: realtà aumentata per veicoli subacquei telecomandati*. Dipartimento di Matematica, Università di Bologna.

November 24, 2000 *Prospettive di ricerca tra Visione e Grafica*. DISI, Università di Genova.

April 30, 1998 *L'algoritmo SMW per il calcolo della disparità stereo*. DISI, Università di Genova.

July 2, 1998 *Robust Feature Tracking*. California Institute of Technology, Pasadena (USA).

November 17, 1997 *Rectification of Stereo Pairs*. Heriot-Watt University, Edinburgh (UK).

February 15, 1997 *Un algoritmo per il calcolo della disparità stereo*. IRST, Trento.

December 18, 1996 *A Symmetry Based Stereo Algorithm*. Heriot-Watt University, Edinburgh (UK).

Invited conference talks

September 14, 2018 **The Long March of 3D Reconstruction: From Tabletop to Outer Space**. ECCV Workshop: 3D Reconstruction in the Wild, Munich (**Invited talk**).

June 3–7, 2018 **Global Network Orientation by Synchronization**. ISPRS Technical Commission II Symposium, Riva del Garda (**Tutorial**).

January 10, 2018 **Automatic three-dimensional modelling from images**. Open Conference: Ai Confini del disegno. Esperienze di modelli virtuali e spazi immersivi, Gorizia (**Invited talk**).

September 7–11, 2015 **Structure from motion: historical overview and recent trends**. 18th International Conference on Image Analysis and Processing, Genoa (**Tutorial**).

June 23–25, 2014 **Structure from motion: state of the art, examples and open issues**. ISPRS Technical Commission V Symposium, Riva del Garda (**Keynote speech**).

June 5, 2013 **Detecting multiple structures in geometric data: the J-linkage approach**. Big Data: una sfida ricca di opportunità, Udine (**Invited talk**).

June 23, 2009 **From structure from motion to videosynthesis**. Convegno Annuale del Gruppo nazionale Telecomunicazioni e Teoria dell'Informazione (GTTI), Parma (**Invited talk**).

September 12, 2006 **Elements of Geometric Computer Vision**. 28th Annual Symposium of the German Association for Pattern Recognition, Berlin (**Tutorial**).

Numerous oral and poster presentations at international conferences.

Research project leadership

2021–2022 **POC-MISE**: Patent valorisation through Proof of Concept projects. Funding: EUR 33,000. Role: **Principal investigator**.

2017–2018 **SPATA**: Automatic profilometry system for monitoring transportation in port areas, Autonomous Region of Friuli Venezia Giulia, POR FESR 2014–2020. Funding: EUR 45,421. Role: **Research unit leader**. Research and development of a laser-scanner-based system for measuring vehicle profiles.

2009 **LIRE**: *Leveraging on Intensity images for Reverse Engineering*, University of Verona, Joint Project 2009. Funding: EUR 65,000. Role: **Scientific coordinator**. Improving 3D model acquisition from laser scanners by exploiting intensity images.

2006 **3-SHIRT**: Three-Dimensional Shape Indexing and Retrieval Techniques, Italian Ministry of Education, University and Research, PRIN 2006. Funding: EUR 73,000. Role: **National scientific coordinator**. Content-based 3D shape retrieval: analysis and segmentation, descriptor design, indexing and matching, and evaluation.

2004 **CooperInt 2004**, University of Verona. Funding: EUR 9,600. Development of international collaborations.

2000 **RIADI**: Retrieval, Integration and Analysis of Hypermedia Data, University of Verona. Young Researchers Programme. Hypermedia information retrieval through data analysis, query languages and their abstract interpretation.

Participation in research projects

2008–2010 **SAMURAI**: *Suspicious and Abnormal behavior Monitoring Using a netwoRk of cAmeras for sItuation awareness*, European Commission, FP7-SEC-2007-01, 217899. Role: **Participant**. Development of a global behaviour-monitoring model using video and non-visual data from fixed and mobile sensors in a distributed network.

2004–2005 **LIMA3D**: Low-cost automatic system for 3D imaging and modelling, Italian Ministry of Education, University and Research, COFIN 2004. Role: **Proposal co-author and participant**. Scientific coordinator: V. Murino. Development of a low-cost system for 3D modelling of

real objects and environments, extracting models from raw sensor data for virtual reality applications.

2001–2003 **ARROV**: *Augmented Reality for Remotely Operated Vehicles based on 3D acoustical and optical sensors for underwater inspection and survey*, European Commission, GROWTH GRD1-2000-25409. Role: **Proposal co-author and participant**. Improving remotely operated underwater vehicles through augmented reality interfaces and 3D models built from acoustic and optical sensors.

2001–2003 **AUREA**: *Augmented Reality for Teleoperation of Free Flying Robots*, Italian Space Agency (ASI), Ricerca Scientifica 2000, 391. Role: **Initiator, proposal co-author and participant**. Augmented reality interface for supervising free-flying service robots: video-based localisation and virtual visual and acoustic feedback overlaid on real feedback.

2001–2002 **SoL**: *The Sounding Landscape*, Hewlett Packard, HP Philanthropic Programme. Role: **Proposal co-author and participant**. Visual analysis and sound synthesis to enable blind users to hear a scene.

2001–2002 **SPADA**: Representation and Processing of Spatial Data in Geographic Information Systems, Italian Ministry of University and Scientific and Technological Research, COFIN 2000. Role: **Proposal co-author**. Scientific coordinator: L. De Floriani. Processing, analysis and automatic annotation of raster maps; extraction and classification of features at different resolutions.

1996–1999 **VENICE**: *Virtual ENvironment Interface by sensory integration for inspection and manipulation Control in multifunctional underwater vehicles*, European Commission, BRITE-EURAM III, BRPR-CT96-0150. Role: **Participant**. Integration of acoustic and optical data to generate a virtual environment enhancing perception and operational capabilities for remotely operated underwater vehicle operators.

Technology transfer

2021 **One-year research fellowship funded by** Comark s.r.l., Udine.

2019 **One-year research fellowship funded by** Helica s.r.l., Amaro (UD).

2017 **One-year research fellowship in collaboration with** 3Dflow s.r.l., *Higher education and development (HEaD)*. European Social Fund funding.

Co-founder of 3Dflow s.r.l. Spin-off of the University of Verona until 2011 and the University of Udine until 2016.

Research contracts with companies. EUR 200,000. Approximate total value.

Doctoral scholarship funded by SeeThrough s.r.l., Brusaporto (BG). Cycle 36.

Doctoral scholarship funded by ST Microelectronics, Agrate (MB). Cycle 28.

PUBLICATIONS

Journal papers

1. Federica Arrigoni, Kathlén Kohn, Andrea Fusiello, and Tomas Pajdla. *An Algebraic Geometry Approach to Viewing Graph Solvability*. **IEEE Transactions on Pattern Analysis and Machine Intelligence** 48(7) (2026), 7229–7240.
2. Sofia Baldini, Filippo Stazi, Riccardo Bernardini, Andrea Fusiello, Paolo Gardonio, and Roberto Rinaldo. *Vibration measurement with neuromorphic vision sensors*. **Mechanical Systems and Signal Processing** 249 (2026), 114102.
3. Sadiq Olayiwola Macaulay, Eleonora Maset, Francesco Boscutti, Paolo Cingano, Francesco Trevisan, Giacomo Trotta, Marco Vuerich, and Andrea Fusiello. *Transferring RGB-Pretrained CNNs to Multispectral UAV Imagery for Salt Marsh Vegetation Classification*. **Remote Sensing** 18(4) (2026).
4. Federica Arrigoni, Andrea Fusiello, Romeo Rizzi, Elisa Ricci, and Tomas Pajdla. *Revisiting Viewing Graph Solvability: an Effective Approach Based on Cycle Consistency*. **IEEE Transactions on Pattern Analysis and Machine Intelligence** 47(5) (2025), 3271–3284.
5. Andrea Fusiello and Pier Luca Montessoro. *Distributed Interval Synchronization*. **IEEE Access** 12 (2024), 1700–1705.
6. Sadiq Olayiwola Macaulay, Eleonora Maset, and Andrea Fusiello. *Classification of compressed full-waveform airborne lidar data*. **Remote Sensing Letters** 15(7) (2024), 729–738.
7. Eleonora Maset and Andrea Fusiello. *Scales of oblique photographs updated*. **ISPRS Journal of Photogrammetry and Remote Sensing** 207 (2024), 198–202.
8. Eleonora Maset, Luca Magri, and Andrea Fusiello. *Principled bundle block adjustment with multi-head cameras*. **ISPRS Open Journal of Photogrammetry and Remote Sensing** 11 (2024), 100051.
9. P. Gardonio, G. Guernieri, E. Turco, L. Dal Bo, R. Rinaldo, and A. Fusiello. *Reconstruction of the sound radiation field from flexural vibration measurements with multiple cameras*. **Mechanical Systems and Signal Processing** 195 (2023), 110289.
10. Paolo Gardonio, Roberto Rinaldo, Loris Dal Bo, Roberto Del Sal, Emanuele Turco, and Andrea Fusiello. *Free-field sound radiation measurement with multiple synchronous cameras*. **Measurement** 188 (2022), 110605.
11. Roberto Del Sal, Loris Dal Bo, Emanuele Turco, Andrea Fusiello, Alessandro Zanarini, Roberto Rinaldo, and Paolo Gardonio. *Structural vibration measurement with multiple synchronous cameras*. **Mechanical Systems and Signal Processing** 157 (2021), 107742.
12. Federica Arrigoni and Andrea Fusiello. *Synchronization Problems in Computer Vision with Closed-Form Solutions*. **International Journal of Computer Vision** 128(1) (January 2020), 26–52.
13. E. Maset, L. Scalera, D. Zonta, I.M. Alba, F. Crosilla, and A. Fusiello. *Procrustes analysis for the virtual trial assembly of large-size elements*. **Robotics and Computer-Integrated Manufacturing** 62 (2020), 101885.
14. F. Arrigoni and A. Fusiello. *Bearing-based Network Localizability: A Unifying View*. **IEEE Transactions on Pattern Analysis and Machine Intelligence** 41(9) (2019), 2049–2069.
15. Antonio Canclini, Francesco Malapelle, Marco Marcon, Stefano Tubaro, and Andrea Fusiello. *View-synthesis from uncalibrated cameras and parallel planes*. **Signal Processing: Image Communication** 79 (2019), 40–53.
16. S. Zorzi, E. Maset, A. Fusiello, and F. Crosilla. *Full-Waveform Airborne LiDAR Data Classification Using Convolutional Neural Networks*. **IEEE Transactions on Geoscience and Remote Sensing** 57(10) (2019), 8255–8261.
17. Federica Arrigoni, Beatrice Rossi, Pasqualina Fragneto, and Andrea Fusiello. *Robust Synchronization in SO(3) and SE(3) via Low-rank and Sparse Matrix Decomposition*. **Computer Vision and Image Understanding** 174 (2018), 95–113.
18. Luca Magri and Andrea Fusiello. *Multiple structure recovery with maximum coverage*. **Machine Vision and Applications** 29(1) (2018), 159–173.
19. Tommaso Banelli, Marco Vuano, Federico Fogolari, Andrea Fusiello, Gennaro Esposito, and Alessandra Corazza. *Automation of peak-tracking analysis of stepwise perturbed NMR spectra*. **Journal of Biomolecular NMR** (2017), 1–14.
20. Luca Magri and Andrea Fusiello. *Multiple structure recovery via robust preference analysis*. **Image and Vision Computing** 67 (2017), 1–15.

21. Luca Magri and Andrea Fusiello. *Multiple structure recovery with T-linkage*. **Journal of Visual Communication and Image Representation** 49 (2017), 57–77.
22. E. Maset, F. Crosilla, and A. Fusiello. *Errors-in-Variables Anisotropic Extended Orthogonal Procrustes Analysis*. **IEEE Geoscience and Remote Sensing Letters** 14(1) (2017), 57–61.
23. Federica Arrigoni, Beatrice Rossi, and Andrea Fusiello. *Spectral Synchronization of Multiple Views in SE(3)*. **SIAM Journal on Imaging Sciences** 9(4) (2016), 1963–1990.
24. Francesco Malapelle, Andrea Fusiello, Beatrice Rossi, and Pasqualina Fragneto. *A data-fusion approach to motion-stereo*. **Signal Processing: Image Communication** 43 (April 2016), 42–53.
25. Andrea Fusiello and Fabio Crosilla. *Solving Bundle Block Adjustment by Generalized Anisotropic Procrustes Analysis*. **ISPRS Journal of Photogrammetry and Remote Sensing** 102 (April 2015), 209–221.
26. Roberto Toldo, Riccardo Gherardi, Michela Farenzena, and Andrea Fusiello. *Hierarchical structure-and-motion recovery from uncalibrated images*. **Computer Vision and Image Understanding** 140 (November 2015), 127–143.
27. M. Corsini, M. Dellepiane, F. Ganovelli, R. Gherardi, A. Fusiello, and R. Scopigno. *Fully Automatic Registration of Image Sets on Approximate Geometry*. **International Journal of Computer Vision** 102(1-3) (March 2013), 91–111.
28. Said Pertuz, Domenec Puig, Miguel Angel Garcia, and Andrea Fusiello. *Generation of All-in-Focus Images by Noise-Robust Selective Fusion of Limited Depth-of-Field Images*. **IEEE Transactions on Image Processing** 22(3) (March 2013), 1242–1251.
29. Roberto Toldo and Andrea Fusiello. *Image-consistent patches from unstructured points with J-linkage*. **Image and Vision Computing** 31(10) (2013), 756–770.
30. A. Fusiello and L. Irsara. *Quasi-Euclidean epipolar rectification of uncalibrated images*. **Machine Vision and Applications** 22(4) (2011), 663–670.
31. A. Colombari and A. Fusiello. *Patch-Based Background Initialization in Heavily Cluttered Video*. **IEEE Transactions on Image Processing** 19(4) (April 2010), 926–933.
32. R. Toldo, U. Castelllani, and A. Fusiello. *The bag of words approach for retrieval and categorization of 3D objects*. **The Visual Computer** 26(10) (October 2010), 1257–1268.
33. Michela Farenzena and Andrea Fusiello. *Stabilizing 3D modeling with geometric constraints propagation*. **Computer Vision and Image Understanding** 113(11) (2009), 1147–1157.
34. U. Castellani, R. Gherardi, A. Fusiello, and V. Murino. *Automatic selection of MRF control parameters by Reactive Tabu Search*. **Image and Vision Computing** 25(11) (2007), 1824–1832.
35. A. Colombari, A. Fusiello, and V. Murino. *Segmentation and Tracking of Multiple Video Objects*. **Pattern Reconition** 40(4) (2007), 1307–1317.
36. A. Fusiello. *A matter of notation: Several uses of the Kronecker product in 3-D computer vision*. **Pattern Recognition Letters** 28(15) (2007), 2127–2132.
37. A. Fusiello. *Specifying Virtual Cameras in Uncalibrated View Synthesis*. **IEEE Transactions on Circuits and Systems for Video Technology** 17(5) (2007), 604–611.
38. U. Castellani, A. Fusiello, V. Murino, L. Papaleo, E. Puppo, and M. Pittore. *A complete system for on-line 3D modelling from acoustic images*. **Signal Processing - Image Communication** 20(9-10) (2005), 832–852.
39. A. Fusiello, M. Farenzena, A. Busti, and A. Benedetti. *Computing Rigorous Bounds to the Accuracy of Calibrated Stereo Reconstruction*. **IEE Proceedings. Vision, Image and Signal Processing** 152(6) (December 2005), 695–701.
40. A. Fusiello, A. Benedetti, M. Farenzena, and A. Busti. *Globally Convergent Autocalibration using Interval Analysis*. **IEEE Transactions on Pattern Analysis and Machine Intelligence** 26(12) (December 2004), 1633–1638.
41. A. Fusiello and V. Murino. *Augmented Scene Modeling and Visualization by Optical and Acoustic Sensor Integration*. **IEEE Transactions on Visualization and Computer Graphics** 10(6) (December 2004), 625–636.
42. U. Castellani, A. Fusiello, and V. Murino. *Registration of Multiple Acoustic Range Views for Underwater Scene Reconstruction*. **Computer Vision and Image Understanding** 87(3) (2002), 78–89.
43. F. Odone, A. Fusiello, and E. Trucco. *Layered Representation of a Video Shot with Mosaicing*. **Pattern Analysis and Applications** 5(3) (2002), 296–305.
44. A. Fusiello. *Uncalibrated Euclidean Reconstruction: A Review*. **Image and Vision Computing** 18(6-7) (May 2000), 555–563.

45. A. Fusiello, V. Roberto, and E. Trucco. *Symmetric Stereo with Multiple Windowing*. **International Journal of Pattern Recognition and Artificial Intelligence** 14(8) (2000), 1053–1066.
46. A. Fusiello, E. Trucco, and A. Verri. *A Compact Algorithm for Rectification of Stereo Pairs*. **Machine Vision and Applications** 12(1) (2000), 16–22.
47. A. Fusiello, E. Trucco, T. Tommasini, and V. Roberto. *Improving Feature Tracking with Robust Statistics*. **Pattern Analysis and Applications** 2(4) (1999), 312–320.
48. E. Trucco, A. Fusiello, and V. Roberto. *Robust Motion and Correspondence of Noisy 3-D Point Sets with Missing Data*. **Pattern Recognition Letters** 20(9) (September 1999), 889–898.
49. A. A. Bertossi and A. Fusiello. *Rate-Monotonic scheduling for hard-real-time systems*. **European Journal of Operational Research** 96(3) (February 1997), 429–443.
50. A. Fusiello and B. Caprile. *Synthesis of Indoor Maps in Presence of Uncertainty*. **Robotics and Autonomous Systems** 22(2) (November 1997), 103–114.

Books

51. Andrea Fusiello. *Computer Vision: Three-dimensional Reconstruction Techniques*. Springer, 2024. isbn: 978-3-031-34506-7. doi: 10.1007/978-3-031-34507-4.
52. Fabio Crosilla, Alberto Beinat, Andrea Fusiello, Eleonora Maset, and Domenico Visintini. *Advanced Procrustes Analysis Models in Photogrammetric Computer Vision*. Vol. 590. CISM International Centre for Mechanical Sciences. Springer, Cham, 2019. isbn: 978-3-030-11759-7. doi: 10.1007/978-3-030-11760-3.
53. Andrea Fusiello. *Visione computazionale. Tecniche di ricostruzione tridimensionale*. II. Franco Angeli, Milano, 2018. isbn: 9788891771933.

Editorships

54. *Guest Editorial: Special Issue on Traditional Computer Vision in the Age of Deep Learning*. Vol. 132. 9. September 2024, pp. 4128–4130. doi: 10.1007/s11263-024-02062-2.
55. Gian Luca Foresti, Andrea Fusiello, and Edwin R. Hancock, eds. *Image Analysis and Processing - ICIAP 2023 - Proceedings, Part I-II*. Vol. 14233. Lecture Notes in Computer Science. Springer, 2023. isbn: 978-3-031-43147-0. doi: 10.1007/978-3-031-43148-7.
56. Adrien Bartoli and Andrea Fusiello, eds. *Computer Vision - ECCV 2020 Workshops Proceedings, Part I-VI*. Vol. 12535-12540. Lecture Notes in Computer Science. Springer, 2020. isbn: 978-3-030-66414-5. doi: 10.1007/978-3-030-66415-2.
57. Andrea Fusiello and Oliver Bimber, eds. *Eurographics 2019 - Posters*. The Eurographics Association, 2019. doi: 10.2312/egp.20192004.
58. Andrea Fusiello, Vittorio Murino, and Rita Cucchiara, eds. *Computer Vision - ECCV 2012. Workshops and Demonstrations Proceedings, Part I-III*. Vol. 7583-7585. Lecture Notes in Computer Science. Springer, 2012. isbn: 978-3-642-33862-5. doi: 10.1007/978-3-642-33863-2.
59. Andrea Fusiello and Michael Wimmer, eds. *Eugraphics 2012 - Posters*. Eurographics Association, 2012.
60. Hongbin Zha, H. Saito, V. Murino, and A. Fusiello, eds. *IEEE Transactions on Systems, Man and Cybernetics, Part B*. Vol. 33. 4. Special issue on 3-D image analysis and modeling. August 2003, pp. 550–553. doi: 10.1109/TSMCB.2003.814779.

Book chapters

61. Fabio Crosilla, Eleonora Maset, and Andrea Fusiello. “Procrustean Photogrammetry: From Exterior Orientation to Bundle Adjustment”. In: *New Advanced GNSS and 3D Spatial Techniques: Applications to Civil and Environmental Engineering, Geophysics, Architecture, Archeology and Cultural Heritage*. Ed. by R. Cefalo, J.B. Zieliński, and M. Barbarella. Lecture Notes in Geoinformation and Cartography. Springer International Publishing, 2018, pp.157–165. isbn: 978-3-319-56218-6. doi: 10.1007/978-3-319-56218-6_12.
62. Andrea Fusiello. “Lemmi vari”. In: *Dizionario di Informatica ICT e media digitali*. Ed. by Edoardo Fleischner. La Piccola Treccani, Scienza e Tecnica. Treccani, Roma, 2012.

63. S. Ivekovic, A. Fusiello, and E. Trucco. “Fundamentals of Multiple-View Geometry”. In: *3D Videocommunication. Algorithms, concepts and real-time systems in human centered communication*. Ed. by O. Schreer, P. Kauff, and T. Sikora. John Wiley & Sons, 2005. Chap. 6. isbn: 9780470022719. doi: 10.1002/0470022736.ch6.

64. A. Trucco, M. Palmese, A. Fusiello, and V. Murino. “Three-dimensional underwater acoustical imaging and processing”. In: *Underwater Acoustic Digital Signal Processing and Communication Systems*. Ed. by R.S.H. Istepanian and M. Stojanovic. Kluwer Academic, 2002, pp.247–274. isbn: 978-0-7923-7304-9.

Selected conferences

65. Federica Arrigoni, Andrea Fusiello, and Tomas Pajdla. *A Direct Approach to Viewing Graph Solvability*. In: **Proceedings of the 18th European Conference on Computer Vision (ECCV)**. (Oral, acceptance rate: 2.3%). Springer Nature Switzerland, 2024, pp.319–334. isbn: 978-3-031-73232-4. doi: 10.1007/978-3-031-73232-4_18.

66. Rakshith Madhavan, Andrea Fusiello, and Federica Arrigoni. *Synchronization of Projective Transformations*. In: **Proceedings of the 18th European Conference on Computer Vision (ECCV)**. Acceptance rate: 27.9%). Springer Nature Switzerland, 2024, pp.18–36. isbn: 978-3-031-72913-3. doi: 10.1007/978-3-031-72913-3_2.

67. Federica Arrigoni, Tomas Pajdla, and Andrea Fusiello. *Viewing Graph Solvability in Practice*. In: **Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)**. (Oral, acceptance rate: 2.4%. Shortlisted for best paper 17/2161). October 2023, pp.8147–8155. doi: 10.1109/ICCV51070.2023.00748.

68. GK Tejus, Giacomo Zara, Paolo Rota, Andrea Fusiello, Elisa Ricci, and Federica Arrigoni. *Rotation Synchronization via Deep Matrix Factorization*. In: **2023 IEEE International Conference on Robotics and Automation (ICRA)**. 2023, pp.2113–2119. doi: 10.1109/ICRA48891.2023.10160548.

69. Andrea Fusiello. *Exact Affine Histogram Matching by Cumulants Transformation*. In: **Proceedings of the International Conference on Image Analysis and Processing (ICIAP)** (May 23–27, 2022). Lecture Notes in Computer Science. (Acceptance rate: 54%.) Springer-Verlag, 2022, pp.199–210. isbn: 978-3-031-06429-6. doi: 10.1007/978-3-031-06430-2_17.

70. Federica Arrigoni, Andrea Fusiello, Elisa Ricci, and Tomas Pajdla. *Viewing Graph Solvability via Cycle Consistency*. In: **Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)** (October 11–17, 2021). (Oral, acceptance rate: 3.4%. Honorable mention award). 2021, pp.5540–5549. doi: 10.1109/ICCV48922.2021.00549.

71. Andrea Porfiri Dal Cin, Luca Magri, Federica Arrigoni, Andrea Fusiello, and Giacomo Boracchi. *Synchronization of Group-Labelled Multi-Graphs*. In: **Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)** (October 11–17, 2021). (Acceptance rate: 26%.) 2021, pp.6453–6463. doi: 10.1109/ICCV48922.2021.00639.

72. Marco Patané and Andrea Fusiello. *Vehicle Classification from Profile Measures*. In: **Proceedings of the 25th International Conference on Pattern Recognition (ICPR)** (January 10–15, 2021). (Oral, acceptance rate: 4.4%). IEEE Computer Society Press, 2021, pp.6656–6663. doi: 10.1109/ICPR48806.2021.9413185.

73. E. Maset, L. Magri, I. Toschi, and A. Fusiello. *Bundle Block Adjustment With Constrained Relative Orientations*. In: **ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences** (August 31–September 2, 2020). Vol. V-2-2020. (Acceptance rate: 19%. Best student paper award). 2020, pp.49–55. doi: 10.5194/isprs-annals-V-2-2020-49-2020.

74. Luca Magri and Andrea Fusiello. *Fitting Multiple Heterogeneous Models by Multi-Class Cascaded T-Linkage*. In: **2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)** (June 16–20, 2019). (Acceptance rate: 25.1%). 2019, pp.7452–7460. doi: 10.1109/CVPR.2019.00764.

75. L. Magri and A. Fusiello. *Reconstruction of interior walls from point cloud data with min-hashed J-linkage*. In: **Proceedings of the International Conference on 3D Vision (3DV)** (September 5–8, 2018). (Acceptance rate: 48.8%). 2018, pp.131–139. doi: 10.1109/3DV.2018.00025.

76. Federica Arrigoni, Eleonora Maset, and Andrea Fusiello. *Synchronization in the Symmetric Inverse Semigroup*. In: **Proceedings of the International Conference on Image Analysis and Processing (ICIAP)** (September 11–15, 2017). Lecture Notes in Computer Science. (Acceptance rate: 60%). Springer International Publishing, 2017, pp.70–81. doi: 10.1007/978-3-319-68548-9_7.

77. Eleonora Maset, Federica Arrigoni, and Andrea Fusiello. *Practical and Efficient Multi-View Matching*. In: **The IEEE International Conference on Computer Vision (ICCV)** (October 22, 2017). (Spotlight acceptance rate: 4.7%). 2017, pp.4568–4576. doi: 10.1109/ICCV.2017.489.

78. Federica Arrigoni, Andrea Fusiello, and Beatrice Rossi. *Camera Motion from Group Synchronization*. In: **Proceedings of the International Conference on 3D Vision (3DV)** (October 25–28, 2016). (Acceptance rate: 44.4%). IEEE, 2016, pp.546–555. doi: 10.1109/3DV.2016.64.

79. Federica Arrigoni, Andrea Fusiello, and Beatrice Rossi. *Global Registration of 3D Point Sets via LRS decomposition*. In: **Proceedings of the 14th European Conference on Computer Vision (ECCV)** (October 8–16, 2016). (Acceptance rate: 26.6%). 2016, pp.489–504. doi: 10.1007/978-3-319-46493-0_30.

80. A. Fusiello and F. Crosilla. *Fast and Resistant Procrustean Bundle Adjustment*. In: **ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences** (July 12–19, 2016). Vol. III. (Acceptance rate: 54%). 2016, pp.35–41. doi: doi:10.5194/isprs-annals-III-3-35-2016.

81. Luca Magri and Andrea Fusiello. *Multiple Models Fitting as a Set Coverage Problem*. In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 27–July 1, 2016). (Acceptance rate: 30%). 2016, pp.3318–3326. doi: 10.1109/CVPR.2016.361.

82. Federica Arrigoni, Andrea Fusiello, and Beatrice Rossi. *On Computing the Translations Norm in the Epipolar Graph*. In: **Proceedings of the International Conference on 3D Vision (3DV)** (October 19–22, 2015). (Acceptance rate: 46%). IEEE, 2015, pp.300–308. doi: 10.1109/3DV.2015.41.

83. Andrea Fusiello, Fabio Crosilla, and Francesco Malapelle. *Procrustean point-line registration and the NPnP problem*. In: **Proceedings of the International Conference on 3D Vision (3DV)** (October 19–22, 2015). (Oral, acceptance rate: 15%). IEEE, 2015, pp.250–255. doi: 10.1109/3DV.2015.35.

84. Luca Magri and Andrea Fusiello. *Robust Multiple Model Fitting with Preference Analysis and Low-rank Approximation*. In: **Proceedings of the British Machine Vision Conference (BMVC)** (September 7–10, 2015). Ed. by Xianghua Xie, Mark W. Jones, and Gary K. L. Tam. (Acceptance rate: 33%). BMVA Press, 2015, pp.20.1–20.12. isbn: 1-901725-53-7. doi: 10.5244/C.29.20.

85. F. Arrigoni, L. Magri, B. Rossi, P. Fragneto, and A. Fusiello. *Robust Absolute Rotation Estimation via Low-rank and Sparse Matrix Decomposition*. In: **Proceedings of the International Conference on 3D Vision (3DV)** (December 8–11, 2014). (Oral, acceptance rate: 15.5%). 2014, pp.491–498. doi: 10.1109/3DV.2014.48.

86. Luca Magri and Andrea Fusiello. *T-Linkage: A Continuous Relaxation of J-Linkage for Multi-Model Fitting*. In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 24–27, 2014). (Acceptance rate: 29.88%). 2014, pp.3954–3961. doi: 10.1109/CVPR.2014.505.

87. S. Fantoni, U. Castellani, and A. Fusiello. *Accurate and automatic alignment of range surfaces*. In: **Proceedings of the 2nd Joint 3DIM/3DPVT Conference: 3D Imaging, Modeling, Processing, Visualization and Transmission (3DIMPVT)** (October 13–15, 2012). (Oral, acceptance rate: 18%). 2012, pp.73–80. doi: 10.1109/3DIMPVT.2012.63.

88. P. Fragneto, A. Fusiello, L. Magri, B. Rossi, and M. Ruffini. *Uncalibrated View Synthesis with Homography Interpolation*. In: **Proceedings of the 2nd Joint 3DIM/3DPVT Conference: 3D Imaging, Modeling, Processing, Visualization and Transmission (3DIMPVT)** (October 13–15, 2012). (Acceptance rate: 70%). 2012, pp.270–277. doi: 10.1109/3DIMPVT.2012.39.

89. V. Garro, F. Crosilla, and A. Fusiello. *Solving the PnP Problem with Anisotropic Orthogonal Procrustes Analysis*. In: **Proceedings of the 2nd Joint 3DIM/3DPVT Conference: 3D Imaging, Modeling, Processing, Visualization and Transmission (3DIMPVT)** (October 13–15, 2012). (Acceptance rate: 70%). 2012, pp.262–269. doi: 10.1109/3DIMPVT.2012.40.

90. Riccardo Gherardi, Michela Farenzena, and Andrea Fusiello. *Improving the Efficiency of Hierarchical Structure-and-Motion*. In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 13–18, 2010). (Acceptance rate: 27%). 2010, pp.1594–1600. doi: 10.1109/CVPR.2010.5539782.

91. Riccardo Gherardi and Andrea Fusiello. *Practical Autocalibration*. In: **Proceedings of the European Conference on Computer Vision (ECCV)** (September 5–11, 2010). Lecture Notes in Computer Science. (Acceptance rate: 28%). Springer, 2010, pp.790–801. doi: 10.1007/978-3-642-15549-9_57.

92. Roberto Toldo and Andrea Fusiello. *Photo-consistent Planar Patches from Unstructured Cloud of Points*. In: **Proceedings of the European Conference on Computer Vision (ECCV)** (September 5–11, 2010). Lecture Notes in Computer Science. (Acceptance rate: 28%). Springer, 2010, pp.589–602. doi: 10.1007/978-3-642-15555-0_43.

93. M. Farenzena, A. Fusiello, and R. Gherardi. *Structure-and-motion Pipeline on a Hierarchical Cluster Tree*. In: **Proceedings of the IEEE International Workshop on 3-D Digital Imaging and Modeling (3DIM)** (October 3–4, 2009). 2009, pp.1489–1496. isbn: 978-1-4244-4441-0. doi: 10.1109/ICCVW.2009.5457435.

94. Roberto Toldo and Andrea Fusiello. *Automatic Estimation of the Inlier Threshold in Robust Multiple Structures Fitting*. In: **Proceedings of the 15th International Conference on Image Analysis and Processing (ICIAP)** (September 8–11, 2009). Vol. 5716. Lecture Notes in Computer Science. (Oral, acceptance rate: 21%. Best student paper award). Springer, 2009, pp.123–131. isbn: 978-3-642-04145-7. doi: 10.1007/978-3-642-04146-4_15.

95. M. Farenzena and A. Fusiello. *3D Surface Models by Geometric Constraints Propagation*. In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 24–26, 2008). (Acceptance rate: 32%). IEEE Computer Society, 2008, pp.1–8. doi: 10.1109/CVPR.2008.4587776.

96. A. Fusiello and L. Irsara. *Quasi-Euclidean Uncalibrated Epipolar Rectification*. In: **Proceedings of the 19th International Conference on Pattern Recognition (ICPR)** (December 8–11, 2008). (Acceptance rate: 62%). 2008, pp.1–4. doi: 10.1109/ICPR.2008.4761561.

97. R. Toldo and A. Fusiello. *Robust Multiple Structures Estimation with J-Linkage*. In: **Proceedings of the European Conference on Computer Vision (ECCV)** (October 12–18, 2008). Vol. 5302. Lecture Notes in Computer Science. (Acceptance rate: 29%). Springer, 2008, pp.537–547. doi: 10.1007/978-3-540-88682-2_41.

98. A. Colombari, A. Fusiello, and V. Murino. *Video Objects Segmentation by Robust Background Modeling*. In: **Proceedings of the International Conference on Image Analysis and Processing (ICIAP)** (September 10–14, 2007). (Oral, acceptance rate: 20%). IEEE Computer Society, 2007, pp.155–164. doi: 10.1109/ICIAP.2007.4362773.

99. M. Farenzena and A. Fusiello. *Recovering Intrinsic Images using an Illumination Invariant Image*. In: **Proceedings of the IEEE International Conference on Image Processing (ICIP)** (September 16–19, 2007). Vol. 3. (Acceptance rate: 49%). 2007, pp.485–488. doi: 10.1109/ICIP.2007.4379352.

100. A. Fusiello and L. Irsara. *An Uncalibrated View-Synthesis Pipeline*. In: **Proceedings of the International Conference on Image Analysis and Processing (ICIAP)** (September 10–14, 2007). (Oral, acceptance rate: 20%). IEEE Computer Society, 2007, pp.609–614. doi: 10.1109/ICIAP.2007.4362844.

101. M. Farenzena, A. Fusiello, and A. Dovier. *Reconstruction with Interval Constraints Propagation*. In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 17–22, 2006). (Acceptance rate: 28%). IEEE Computer Society, 2006, pp.1185–1190. doi: 10.1109/CVPR.2006.246.

102. A. Colombari, A. Fusiello, and V. Murino. *Uncalibrated Interpolation of Rigid Displacements for View Synthesis*. In: **Proceedings of the IEEE International Conference on Image processing (ICIP)** (September 11–14, 2005). (Acceptance rate: 46%). 2005, pp.1049–1052. doi: 10.1109/ICIP.2005.1529934.

103. M. Farenzena, A. Busti, A. Fusiello, and A. Benedetti. *Rigorous accuracy bounds for calibrated stereo reconstruction*. In: **Proceedings of the 17th International Conference on Patter Recognition (ICPR)** (August 23–26, 2004). Vol. 4. (Oral, acceptance rate: 18%). IEEE Computer Society, 2004, pp.288–292. doi: 10.1109/ICPR.2004.1333760.

104. R. Marzotto, A. Fusiello, and V. Murino. *High Resolution Video Mosaicing with Global Alignment*. In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 27–July 2, 2004). Vol. I. (Acceptance rate: 30%). IEEE Computer Society, 2004, pp.692–698. doi: 10.1109/CVPR.2004.1315099.

105. A. Sanson, U. Castellani, A. Fusiello, and V. Murino. *Fast Model Tracking with Multiple Cameras for Augmented Reality*. In: **Proceedings of the ACM Symposium on Virtual Reality Software and Technology (VRST)** (November 10–12, 2004). (Acceptance rate: 39%). 2004, pp.170–173. doi: 10.1145/1077534.1077569.

106. A. Benedetti, A. Busti, M. Farenzena, and A. Fusiello. *Globally Convergent Autocalibration*. In: **Proceedings of the 9th IEEE International Conference on Computer Vision (ICCV)** (October 14–17, 2003). Vol. II. (Acceptance rate: 21%). IEEE Computer Society, 2003, pp.1426–1432. doi: 10.1109/ICCV.2003.1238657.

107. A. Fusiello, M. Aprile, R. Marzotto, and V. Murino. *Mosaic of a video shot with multiple moving objects*. In: **Proceedings of IEEE International Conference on Image Processing (ICIP)** (September 14–18, 2003). Vol. II. (Acceptance rate: 43%). 2003, pp.307–310. doi: 10.1109/ICIP.2003.1246678.

108. A. Fusiello, S. Caldrer, S. Ceglie, N. Mattern, and V. Murino. *View Synthesis from Uncalibrated Images Using Parallax*. In: **Proceedings of the 12th International Conference on Image Analysis and Processing (ICIAP)** (September 17–19, 2003). (Oral, acceptance rate: 26%). IEEE Computer Society, 2003, pp.146–151. doi: 10.1109/ICIAP.2003.1234041.

109. A. Fusiello, U. Castellani, L. Ronchetti, and V. Murino. *Model Acquisition by Registration of Multiple Acoustic Range Views*. In: **Proceedings of the 7th European Conference on Computer Vision (ECCV)**. Ed. by A. Heyden, G. Sparr, M. Nielsen, and P. Johansen. Vol. 2351. Lecture Notes in Computer Science. (Acceptance rate: 38%). Springer, 2002, pp.805–819. doi: 10.1007/3-540-47967-8_54.

110. A. Fusiello, A. Panuccio, V. Murino, F. Fontana, and D. Rocchesso. *A Multimodal Electronic Travel Aid Device*. In: **Proceedings of the Fourth IEEE International Conference on Multimodal Interfaces (ICMI)** (October 14–16, 2002). (Acceptance rate: 43%). IEEE Computer Society, 2002, pp.39–44. doi: 10.1109/ICMI.2002.1166966.

111. V. Murino, L. Ronchetti, U. Castellani, and A. Fusiello. *Reconstruction of Complex Environments by Robust Pre-aligned ICP*. In: **Proceedings of the International Conference on 3-D Digital Imaging and Modeling (3DIM)** (May 28–June 1, 2001). IEEE Computer Society Press, 2001, pp.187–194. doi: 10.1109/IM.2001.924432.

112. A. Valinetti, A. Fusiello, and V. Murino. *Model tracking for Video-based Virtual Reality*. In: **Proceedings of the 11th International Conference on Image Analysis and Processing (ICIAP)** (September 26–28, 2001). (Oral, acceptance rate: 28%). IEEE Computer Society Press, 2001, pp.372–377. doi: 10.1109/ICIAP.2001.957038.

113. V. Murino, A. Fusiello, N. Iuretigh, and E. Puppo. *3D Mosaicing for Environment Reconstruction*. In: **Proceedings of the International Conference on Pattern Recognition (ICPR)** (September 3–8, 2000). Vol. 3. (Acceptance rate: 68%). IEEE Computer Society Press, 2000, pp.362–366. doi: 10.1109/ICPR.2000.903559.

114. A. Censi, A. Fusiello, and V. Roberto. *Image Stabilization by Features Tracking*. In: **Proceedings of the 10th International Conference on Image Analysis and Processing (ICIAP)** (September 27–29, 1999). (Acceptance rate: 57%). IEEE Computer Society Press, 1999, pp.665–667. doi: 10.1109/ICIAP.1999.797671.

115. A. Fusiello, R. Giannitrapani, V. Isaia, and V. Murino. *Virtual Environment Modeling by Integrated Optical and Acoustic Sensing*. In: **Proceedings of the 2nd International Conference on 3-D Digital Imaging and Modeling (3DIM)** (October 4–8, 1999). IEEE Computer Society Press, 1999, pp.437–446. doi: 10.1109/IM.1999.805375.

116. T. Tommasini, A. Fusiello, E. Trucco, and V. Roberto. *Making Good Features Track Better*. In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 23–25, 1998). (Acceptance rate: 30%). IEEE Computer Society Press, 1998, pp.178–183. doi: 10.1109/CVPR.1998.698606.

117. A.A. Bertossi, A. Fusiello, and L.V. Mancini. *Fault-Tolerant Deadline-Monotonic Algorithm for Scheduling Hard-Real-Time Tasks*. In: **Proceedings of the 11th International Parallel Processing Symposium** (April 1–5, 1997). (Acceptance rate: 38%). IEEE Computer Society Press, 1997, pp.133–138. doi: 10.1109/IPPS.1997.580868.

118. A. Fusiello, V. Roberto, and E. Trucco. *Efficient Stereo with Multiple Windowing*. In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 17–19, 1997). (Acceptance rate: 32%). IEEE Computer Society Press, 1997, pp.858–863. doi: 10.1109/CVPR.1997.609428.

119. A. Fusiello, E. Trucco, and A. Verri. *Rectification with Unconstrained Stereo Geometry*. In: **Proceedings of the British Machine Vision Conference (BMVC)** (September 8–11, 1997). Ed. by A. F. Clark. (Acceptance rate: 62%). BMVA Press, GB, 1997, pp.400–409.

Workshops, minor conferences and national conferences are omitted. Complete publication list.

Google Scholar – July 2022	Citations	h-index	i10-index
All	7423	40	105
Since 2017	2402	24	58

Updated bibliometric data (VQR, ASN): bibliometric profile.

Patents

February 2019 — Yash Singh, Roberto Toldo, Luca Magri, Simone Fantoni, and Andrea Fusiello. *Method for 3D modelling based on structure from motion processing of sparse 2D images*. U.S. Patent US 10,198,858 B2. February 2019.

15 Maggio 2018 — Fabio Crosilla, Andrea Fusiello, Eleonora Maset, and Stefano Zorzi. *Apparato e Metodo di Classificazione di Dati a Forma d'onda Completa da Segnali Retroriflessi*. IT102018000005375. 15 Maggio 2018.

Awards and distinctions

Since 2021 — **IEEE Senior Member**.

2021 — *Marr Prize Best Paper Honorable Mention*, ICCV.

1999–2000 — **Regional Innovation Award**, Veneto Innovazione.

TEACHING EXPERIENCE

Undergraduate teaching

Since 2014/2015 **Fundamentals of Computer Science** (6 ECTS), Bachelor's degree in Mechanical Engineering, University of Udine.

Since 2012/2013 **Fundamentals of Computer Science** (6 ECTS), Bachelor's degree in Civil Engineering, University of Udine.

2011/2012–2012/2013 **Elements of Computer Science** (6 ECTS), Bachelor's degree in Architectural Sciences, University of Udine.

2009/2010–2010/2011 **Computer Graphics** (6 ECTS), Bachelor's degree in Computer Science, Faculty of Mathematical, Physical and Natural Sciences, University of Verona.

2008/2009 **Computational Geometry: Advanced Module** (2 ECTS), Bachelor's degree in Applied Mathematics, Faculty of Mathematical, Physical and Natural Sciences, University of Verona.

2007/2008–2008/2009 **Basic Computing Laboratory** (2 ECTS), Bachelor's degree in Computer Science, Faculty of Mathematical, Physical and Natural Sciences, University of Verona.

2005/2006–2006/2007 **Human-Computer Interaction** (3 ECTS), Bachelor's degree in Communication Science, Faculty of Arts and Philosophy, University of Verona.

2002/2003–2008/2009 **Computer Graphics** (5 ECTS), Bachelor's degree in Multimedia Computer Science (formerly Information Technology), Faculty of Mathematical, Physical and Natural Sciences, University of Verona.

2001/2002 **Programming Laboratory** (4 ECTS), Bachelor's degree in Computer Science, Faculty of Mathematical, Physical and Natural Sciences, University of Verona.

1998/1999 **Integrated Course in Medical Physics, Statistics and Computer Science** (15 hours), Nursing Diploma, University of Udine, Pordenone. Contract lecturer.

1998/1999–2000/2001 **Computing Laboratory accompanying the Programming course** (60 hours), Computer Science degree, Faculty of Mathematical, Physical and Natural Sciences, University of Verona. Contract lecturer.

Graduate teaching

Since 2016/2017 **Digital Archaeology** (6 ECTS), Master's degree in Ancient Civilisations, University of Udine (offered every other year).

Since 2012/2013 **Computer Vision** (6 ECTS), Master's degree in Electronic Engineering, University of Udine. The course title has changed over time.

2009/2010–2010/2011 **Computational Vision** (6 ECTS), Master's degree in Computer Science and Engineering, Faculty of Mathematical, Physical and Natural Sciences, University of Verona.

2004/2005 **Communication and Interaction** (6 ECTS), Master's degree in Electronic and Multimedia Publishing, Faculty of Arts and Philosophy, University of Verona.

2002/2003–2008/2009 **Computational Vision** (5 ECTS), Master's degree in Intelligent and Multimedia Systems, Faculty of Mathematical, Physical and Natural Sciences, University of Verona.

2001/2002 **Computer Vision** (5 ECTS), Computer Science degree, Faculty of Mathematical, Physical and Natural Sciences, University of Verona.

Scuola Superiore (University honours school)

2021/2022 **Advanced Mathematical Methods for Engineering Problems** (14 hours), Scuola Superiore (University honours school), University of Udine.

2013/2014 **Neural Networks** (28 hours), Scuola Superiore (University honours school), University of Udine.

Doctoral teaching

2013–2015 **Lecture**: **Introduction to scientific computing in MATLAB**, PhD programme in Industrial and Information Engineering, Scientific Skills Training Courses, University of Udine.

July 5, 2007 **Seminar**: **Image-based rendering methods**, Summer School in Information Engineering, University of Padua, Brixen.

July 2006 **Lecture series**: **Introduction to Computer Graphics**, Doctoral School in Information Engineering, University of Padua.

June–July 2005 **Lecture series**: **Elements of Multiple View Geometry for Computer Vision**, Doctoral School in Information Engineering, University of Padua.

2003 **Lecture**: **Regularization Machines for Learning**, Modelli per l'analisi e l'elaborazione dei dati, University of Verona.

Advanced teaching

July 8, 2022 **Seminar**: **Structure from motion in computer vision and photogrammetry**, Advanced Professional Training, International Centre for Mechanical Sciences (CISM), Udine.

September 23, 2021 **Photogrammetric computer vision with cultural heritage applications**, VISMAC Summer School, Palermo.

August 31, 2018 **Large-Scale Camera Estimation in 3D Vision: lab session**, VISMAC Summer School, Vico Equense.

April 2–16, 2017 **Lecture series**: **Elements of Photogrammetric Computer Vision**, VIBOT (Erasmus Mundus) Master's programme, IUT Le Creusot – Université de Bourgogne.

2016/2017 **Course**: **Computer Graphics and 3D Reconstruction** (24 hours), Second-level postgraduate programme in Visual Computing and Multimedia Technology, University of Modena and Reggio Emilia.

June 17, 2016 **Lab activity on 3D computer vision**, VISMAC Summer School, Grado.

July 14–20, 2013 **Seminar**: **The devil is in the details: anatomy of a structure-from-motion pipeline**, International Computer Vision Summer School, Calabria.

October 23, 2012 **Introduction to image-based modeling**, VISMAC Summer School, Genoa.

November 11, 2010 **Recent developments in image-based 3D reconstruction**, VISMAC Summer School, Catania.

April 14–16, 2010 **Lecture series**: **School of Photogrammetry, Computer Vision and Laser Scanning: Paradigms and Algorithms**, International Centre for Mechanical Sciences (CISM), Udine.

October 8–29, 2008 **Lecture series**: **Elements of Geometric Computer Vision**, VIBOT (Erasmus Mundus) Master's programme, IUT Le Creusot – Université de Bourgogne.

September 29, 2006 **Camera calibration: problems and algorithms**, VISMAC Summer School, Palermo.

June 2006 **Lecture series**: **Elements of Geometric Computer Vision**, Interactive Digital Multimedia IGERT Programme, University of California, Santa Barbara.

Master's and Laurea thesis supervision

Supervisor of 53 theses in Computer Science (Master's, specialist or pre-reform degrees). https://fusiello.github.io/index.php/Thesis_supervised.html.

Supervisor of 3 Master's theses in Electronic Engineering.

PhD thesis supervised

Abdul Salam Rasmi Asraf Ali. *CBCT Motion Correction by Optimization for Dental Applications*. PhD in Industrial and Information Engineering. University of Udine. Cycle 36.

2019 Eleonora Maset. *Advanced Methods for LiDAR and Photogrammetric Data Processing: from Procrustes Analysis to Deep Learning*. PhD in Industrial and Information Engineering. University of Udine.

2017 Federica Arrigoni. *Synchronization Problems in Computer Vision*. PhD in Industrial and Information Engineering. University of Udine.

2015 Luca Magri. *Stima di modelli ibridi per dati eterogenei*. PhD in Mathematics and Statistics for Computational Sciences. University of Milan.

2015 Francesco Malapelle. *Towards a Complete Pipeline for 3DS Conversion of Monocular Uncalibrated Images*. PhD in Industrial and Information Engineering. University of Udine.

2013 Valeria Garro. *Image localization and parsing using 3D structure*. PhD in Computer Science. Department of Computer Science, University of Verona.

2013 Roberto Toldo. *Towards automatic acquisition of high-level 3D models from images*. PhD in Computer Science. Department of Computer Science, University of Verona.

2010 Riccardo Gherardi. *Advances in 3D reconstruction*. PhD in Computer Science. Department of Computer Science, University of Verona.

2009 Davide Moschini. *Tracking human motion with multiple cameras using articulated ICP with hard constraints*. PhD thesis. Department of Computer Science, University of Verona.

2007 Michela Farenzena. *Geometry and Appearance Modelling from Images*. PhD thesis. Department of Computer Science, University of Verona.

2006 Andrea Colombari. *Image-Based Techniques for View Synthesis and Video Editing*. PhD in Computer Science. Department of Computer Science, University of Verona.

2003 Umberto Castellani. *Image-based modeling: from sensory data to 3D models*. PhD in Computer Science. Department of Computer Science, University of Verona. With V. Murino.

SERVICE

University service

October 2014–September 2019 **Rector's Delegate**, Language and Audiovisual Centre (CLAV), University of Udine.

Since 2013 **Board member**, PhD programme in Industrial and Information Engineering, University of Udine.

2012–2014 **Chair of the Admissions Examination Committee and TOLC coordinator**, Engineering degree programmes, University of Udine.

2010–2011 **Chair of the Teaching Committee**, Master's degree in Computer Science and Engineering, Faculty of Mathematical, Physical and Natural Sciences, University of Verona.

2010–2011 **Vice-President**, Master's degree in Computer Science and Engineering, Faculty of Mathematical, Physical and Natural Sciences, University of Verona.

2009/2010–2010/2011 **Director and initiator**, *Computer Game Development*, University of Verona. First-level postgraduate programme.

2006–2009 **Member**, Teaching Committee, Joint Teaching Committee and Self-Assessment Committee of the Computer Science Degree Programme Board, University of Verona.

2006–2009 **Elected member**, Graduate School of Sciences, Engineering and Medicine, University of Verona.

2006–2009 **Elected member**, External Relations Committee, Faculty of Mathematical, Physical and Natural Sciences, University of Verona.

2005 **Secretary**, Computer Science Degree Programme Board, University of Verona.

2000–2012 **Board member**, PhD programme in Computer Science, University of Verona.

Scientific association boards

2019–2020 **Auditor**, Italian Association for Computer Vision, Pattern Recognition and Machine Learning (CVPL).

June 2016–June 2022 **Co-chair**, *ISPRS WG II/1 (International Society for Photogrammetry and Remote Sensing)*.

2006–2009 **Elected member**, Executive Board of the Italian Chapter of Eurographics.

Editorial boards

Since 2021 **Associate editor**, IEEE Transactions on Image Processing.

October 2018 **Guest editor**, *Selected Papers from the 19th International Conference on Image Analysis and Processing (ICIAP) 2017*, IET Computer Vision. With: F. Stanco, D. Tegolo.

2013–2016 **Associate editor**, IET Computer Vision.

Since 2011 **Scientific board member**, *Informatica*, Franco Angeli. Book series.

2006–2009 **Associate editor**, Pattern Recognition.

August 2003 **Guest editor**, *Special issue on 3-D Image Analysis and Modelling*, IEEE Transactions on System, Man, and Cybernetics. With: Hong Bin Zha, Hideo Saito, Vittorio Murino.

Journal reviewer

Numerous journals, including: *International Journal of Computer Vision*, *IEEE Journal on Oceanic Engineering*, *IEEE Transaction on Pattern Analysis and Machine Intelligence*, *IEEE Transaction on Circuits And Systems For Video Technology*, *IEEE Transactions on Multimedia*, *IEEE Transaction on System, Man, and Cybernetics*, *IEEE Transaction on*

Image Processing, *IEEE Transaction on Medical Imaging*, *Machine Vision and Applications*, *Computer Vision and Image Understanding*, *Image and Vision Computing Journal*, *Pattern Recognition Letters*, *ISPRS Journal on Photogrammetry and Remote Sensing*.

Conference organisation

October 16, 2021 — **Organising committee member**, *1st Workshop on Traditional Computer Vision in the Age of Deep Learning (ICCV Workshops)*.

August 23–28, 2020 — **Workshops Chair**, *European Conference on Computer Vision (ECCV)*, Glasgow (UK). With: A. Bartoli.

September 9–13, 2019 — **Area Chair**, *20th International Conference on Image Analysis and Processing (ICIAP)*, Trento. With: A. Del Bue, F. Tombari.

June 16–17, 2019 — **Organising committee member**, *Photogrammetric Computer Vision Workshop (CVPR Workshops)*, Long Beach (US).

May 6–10, 2019 — **Poster Chair**, *40th Annual Conference of the European Association for Computer Graphics (Eurographics)*, Genoa. With: O. Bimber.

September 5–8, 2018 — **General Chair**, *6th International Conference on 3D Vision (3DV)*, Verona.

September 21–22, 2017 — **Organising committee member**, *ISPRS Workshop: Photogrammetric 3D Reconstruction for Geo-Applications*, Wuhan (China).

September 11–15, 2017 — **Area Chair**, *19th International Conference on Image Analysis and Processing (ICIAP)*, Catania. With: D. Fofi.

September 7–11, 2015 — **Area Chair**, *18th International Conference on Image Analysis and Processing (ICIAP)*, Genoa. With: M. Goesele.

October 7–13, 2012 — **Workshops Chair**, *12th European Conference on Computer Vision (ECCV)*, Florence. With: V. Murino.

May 13–18, 2012 — **Poster Chair**, *33rd Annual Conference of the European Association for Computer Graphics (Eurographics)*, Cagliari. With: M. Wimmer.

October 22–23, 2009 — **Chair**, *7th Eurographics Italian Chapter Conference (EG-IT)*, Verona. With: U. Castellani, A. Giachetti, V. Murino.

September 6–8, 2005 — **Chair**, *Computer Vision for Augmented Reality and Computer Augmented Environments, 13th ICIAP*, Cagliari. With: V. Murino.

June 17–18, 2002 — **Programme Chair**, *Convegno del Gruppo Italiano Ricercatori in Pattern Recognition (GIRPR)*, Verona.

September 21–29, 1999 — **Chair**, *Young Scientist Forum, 10th International Conference on Image Analysis and Processing*, Venice.

Programme committee / Conference reviewer

Workshops and minor conferences omitted.

2006, 2007, 2017, 2022 — *British Machine Vision Conference (BMVC)*.

2007, 2009, 2013, 2021 — *International Conference on Image Analysis and Processing (ICIAP)*.

2014–2017 — *3D Vision Conference (3DV)*.

2008, 2009, 2010 — *International Conference on Image Processing (ICIP)*.

2007, 2009 — *International Conference on Computer Vision/Computer Graphics Collaboration Techniques and Applications (MIRAGE)*.

Since 2006 — *European Conference on Computer Vision (ECCV)*.

Since 2005 (except 2017) *IEEE International Conference on Computer Vision and Pattern Recognition (CVPR)*.

Since 2005 (except 2017) *IEEE International Conference on Computer Vision (ICCV)*.

Organisation of other events

July 8, 2022 **Coordinator**, Recent Developments in Digital Photogrammetry: 3D Acquisition and Automatic Information Extraction Techniques, Advanced Professional Training, International Centre for Mechanical Sciences (CISM), Udine. With: G. Guidi.

June 13–17, 2016 **Organising committee member**, *VISMAC International Summer School*, Grado (GO).

April 14–16, 2010 **Coordinator**, School of Photogrammetry, Computer Vision and Laser Scanning: Paradigms and Algorithms, International Centre for Mechanical Sciences (CISM), Udine. With: F. Crosilla.

2004–2010 **Coordinator**, *VIPS Advanced School on Computer Vision, Pattern Recognition and Image Processing*, Department of Computer Science, University of Verona. With: V. Murino.

Doctoral thesis review and examination

2022 Giang Truong, Edith Cowan University (Australia).

2019 Adel Saleh Ali Alraimi, Universitat Rovira i Virgili.

2019 Bruno Fanini, Sapienza University of Rome.

2018 Andrea Apicella, University of Naples Federico II.

2018 Guido Averna, University of Palermo.

2015 Luca Cosmo, Ca' Foscari University of Venice.

2015 Danda Pani Paudel, Université de Bourgogne.

2014 Dong Han, Politecnico di Milano.

2014 Francesco Michielin, University of Padua.

2013 Hoi Sim Wong, University of Adelaide.

2011 Fabio Bellavia, University of Palermo.

2011 Marco Cipolla, University of Palermo.

2010 Andrea Albarelli, Ca' Foscari University of Venice.

2009 Luca Ballan, University of Padua.

2009 Thierry Molinier, Université de Bourgogne.

2007 Stefano Arca, University of Milan.

2007 Elisabetta Del Ponte, University of Genoa.

2007 Simone Gasparini, Politecnico di Milano.

2007 Pietro Zanuttigh, University of Padua.

2006 Marco Zuliani, University of California, Santa Barbara.

2004 Miguel Arredondo, Heriot-Watt University, Edinburgh.

Research project reviewer

2021 Israel Science Foundation.

2019 Université Bourgogne Franche-Comté.

2007, 2009, 2013 Autonomous Province of Trento.

2011 Futuro in Ricerca (FIRB).

2010 Austrian Science Fund (FWF).

2009 University of Applied Sciences and Arts of Southern Switzerland.

2007 Research Projects of National Interest (PRIN).

2005 Italian Committee for Research Evaluation (CIVR).

October 22, 2026 Andrea Fusiello